MIL-T-52497A(ME)
12 September 1972
SUPERSEDING
MIL-T-52497(MO)
15 February 1966

MILITARY SPECIFICATION

TRANSFORMERS, POWER, ISOLATION: SERIES-CIRCUIT, AIRPORT-LIGHTING, 6.6/6.6 AMPERES

1. SCOPE

1.1 Scope. This specification covers series-circuit, airport-lighting power transformers.

1.2 Classification. Transformers shall be of the following types, as specified (see 6.2):

Type I - 30 to 45 watts
Type II - 200 watts

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-S-571	- Solder; Tin Alloy; Lead-Tin Alloy; and Lead Alloy.

Military

MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-3432	- Cable and Wire, Electrical (Power and Control; Semi-Flexible, Flexible and Extra-Flexible, 300 and 600 Volts).
MIL-I-3930	- Insulating and Jacketing Compounds, Electrical (for Cables, Cords, and Wires).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:08Z
Check the source to verify that this is the current version before use.

MIL-S-6872	- Soldering Process, General Specification for.
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of.
MIL-I-24391	- Insulation Tape, Electrical, Plastic, Pressure-Sensitive.
MIL-C-52286	- Cable Assemblies, Power, Electrical, (With Molded On Terminations).

STANDARDS

Federal

FED. STD. No. 228	- Cable and Wire, Insulated, Methods of Testing.

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-143	- Standards and Specifications, Order of Precedence for the Selection of.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-810	- Environmental Test Methods.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

B 3	- Soft or Annealed Copper Wire.
B 8	- Concentric - Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:08Z
Check the source to verify that this is the current version before use.

B 33 - Tinned Soft or Annealed Copper Wire for Electrical Purposes.

B 189 - Lead-Coated and Lead-Alloy-Coated Soft Copper Wire for Electrical Purposes.

D 1351 - Polyethylene Insulated Wire and Cable.

D 2633 - Testing Thermoplastic Insulated and Jacketed Wire and Cable.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

FEDERAL AVIATION AGENCY

L-823 - Plug and Receptacle, Cable Connectors.

(Application for copies should be addressed to the Federal Aviation Agency, Distribution Section, HQ-438, Washington, DC 20553.)

3. REQUIREMENTS

3.1 Description. The transformers shall consist of primary and secondary coils wound upon a core and enclosed in a waterproof, and watertight case with connectors molded on the primary and secondary leads. A watertight cap shall be provided for each connector.

3.2 First article (preproduction model). The supplier shall furnish one or more transformers for examination and testing within the time frame specifi (see 6.2), to prove prior to starting production that his production methods and choice of design detail will produce transformers that comply with the requirements of this specification. Examination and tests shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.5).

3.3 Materials.

3.3.1 Selection of materials. Materials shall be as specified herein. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification. Specifications and standards for materials not specified herein shall be selected in accordance with MIL-STD-143.

3.3.2 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:08Z
Check the source to verify that this is the currentversion before use.

a varnish conforming to MIL-V-173, with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.

(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.3.3 Metals. Metals shall be inherently corrosion resistant or treated to resist corrosion due to fuels, salt spray, and atmospheric conditions as may be encountered in storage or service.

3.3.3.1 Dissimilar metals. Non-permissible galvanic couples (dissimilar metals) as defined in MIL-T-704 shall be insulated as specified in MIL-T-704.

3.3.4 Solder. Solder, if used for electrical connections, shall have a melting point not less than 25° C. higher than the highest temperature developed in the transformer either during construction, or use, and shall be selected from those specified in QQ-S-571. Soldering, when used, shall be in accordance with MIL-S-6872.

3.4 Construction. The transformers shall be built to withstand without damage (see 6.4) the strains, jars, vibration, and other conditions incident to shipping, storage, installation, and service as specified herein. Transformers shall be similar to figure 1 (see 6.3). The exact shape shall be optional, provided all requirements specified herein are met (see 6.3).

3.4.1 Performance. The transformers shall perform under the environmental, mechanical, and electrical conditions specified herein and be designed for continuous service at ambient temperatures from minus 55° C. to plus 65° C., either on the ground's surface, buried directly in the ground, or submerged in water. The transformers shall withstand storage and transit conditions at temperatures from minus 55° C. to plus 75° C. without damage (see 6.4).

3.4.2 Rating. The transformers shall operate within a frequency range of 50 to 60 Hertz and be furnished in either of the following ratings (see 1.2):

Type I - 30 to 45 watts, 6.6 ampere primary/6.6 ampere secondary.

Type II - 200 watts, 6.6 ampere primary/6.6 ampere secondary.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:08Z
Check the source to verify that this is the currentversion before use.

3.4.3 Windings. The windings shall be completely insulated from each other and the core. The transformers shall operate indefinitely on load, short-circuit, or open-circuit conditions in the secondary with rated current and frequency in the primary. The transformers shall have sufficient capacity to allow for lamp aging and secondary contact losses.

3.4.4 Case. Each transformer shall be enclosed in a hermetic watertight case. No portion of the case shall be less than 1/4 inch thick. Sharp corners and edges of the core and coil assembly shall be eliminated or provisions shall be made so that they will not cut the case if the transformers are dropped or handled roughly. The case shall be so constructed that moisture accidentally getting into the leads or connectors cannot be conducted through the leads into the transformer. The case shall be molded from either rubber or rubber-like compounds and either molded directly on the transformers or premolded and compound filled. Internal air pockets shall be eliminated insofar as is practicable and the assembly shall withstand rough handling and mechanical abuse without damage when tested as specified in 4.5.2.3. The case shall withstand exposure to sun, oil, fungus moisture, and acid and alkaline soils without damage when tested as specified in 4.5.2.4.4 through 4.5.2.4.8. The case shall form a hermetic, watertight junction with the leads described in 3.4.5.

3.4.5 Leads. Each transformer shall be provided with a two-conductor secondary lead and two single-conductor primary leads. The leads shall be fastened to the transformer so that carrying by the leads will not loosen or break the electrical connections or water seal. A connector shall be molded on the outer end of each lead. Lead lengths shall be 30 plus or minus 3 inches for the primary leads and 54 plus or minus 3 inches for the secondary leads.

3.4.5.1 Primary leads. The primary leads shall be 5000-volt, No. 8 AWG, single-conductor cables conforming to ASTM D 1351, except that Methods of Test, paragraph 5, shall be in accordance with ASTM D 2633. The conductor shall be soft or annealed copper wire conforming to ASTM B 3, and stranded in accordance with ASTM B 8, class C. The conductor shall be coated with pure tin in accordance with ASTM B 33 or lead or a lead alloy in accordance with ASTM B 189. The insulated cable shall be covered with a type JS-L or type JN-L heavy-duty jacket in accordance with MIL-I-3930, and MIL-C-3432. The leads shall be marked H1 and H2. The H1 lead shall connect to a male, single-pole connector plug conforming to figure 6a of FAA-L-823, and H2 lead shall connect to a female, single-pole connector plug conforming to figure 6b of FAA-L-823.

3.4.5.2 Secondary lead. The secondary lead shall be a 600-volt, No. 14 AWG two-conductor cable conforming to ASTM D 1351, except that Methods of Test, paragraph 5, shall be in accordance with ASTM D 2633. The conductor and jacket shall conform to the specifications given in 3.4.5.1. The lead shall terminate in a female, two-pole connector plug conforming to figure 1c of FAA-L-823. The connector receptacle shall be wired so that the large contact

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:08Z
Check the source to verify that this is the current version before use.

will connect to the X1 lead of the transformer secondary and the smaller contact will connect to the X2 lead.

3.4.6 Caps. A watertight cap conforming to MIL-C-52286 shall be placed on each connector for protection during shipment and installation. The caps shall protect the mating surfaces of the connector and connector receptacle from moisture and dirt. Insulation tape at least 7 mils thick and 3/4 inch wide, conforming to MIL-I-24391, shall be wrapped over the joining seam to hold the cap in place and to give additional protection from moisture and dirt.

3.5 Performance.

3.5.1 Electrical requirements.

3.5.1.1 When tested as specified in 4.5.2.2.1, the transformer shall conform to the values specified in table I.

3.5.1.2 Insulation resistance. When transformers are tested as specified in 4.5.2.2.2 or 4.5.2.2.3, the insulation resistance value shall be not less than the value specified in table II. There shall be no damage to the transformer.

3.5.2 Mechanical (physical) requirements.

3.5.2.1 Impact. When tested as specified in 4.5.2.3.1, there shall be no damage to the transformer or a change of more than 1 percent in the results obtained in the test specified in 4.5.2.2.1.

3.5.2.2 Vibration. When tested as specified in 4.5.2.3.2, the transformer shall not be damaged by vibration.

3.5.3 Environmental requirements.

3.5.3.1 Temperature rise. When transformers are tested as specified in 4.5.2.4.1, the temperature rise shall not exceed 55° C. when operating under rated load, short-circuited, or open-circuited conditions with rated current in the primary.

3.5.3.2 High temperature. The transformers shall not be damaged when tested as specified in 4.5.2.4.2.

3.5.3.3 Low temperature. The transformers shall not be damaged when tested as specified in 4.5.2.4.3.

3.5.3.4 Fungus resistance. The transformers shall not be damaged when tested as specified in 4.5.2.4.6.

3.5.3.5 Sunshine. The transformers shall not be damaged when tested as specified in 4.5.2.4.4.

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:08Z
Check the source to verify that this is the current version before use.

3.5.3.6 Oil resistance. The transformers shall not be damaged when tested as specified in 4.5.2.4.7.

3.5.3.7 Moisture resistance. The transformers shall not be damaged when tested as specified in 4.5.2.4.5.

3.5.3.8 Acid and alkali resistance. The transformers shall not be damaged when tested as specified in 4.5.2.4.8.

3.6 Weight. The weights of the transformers shall not exceed 10 pounds for the type I transformer or 15 pounds for the type II transformer.

3.7 Identification marking. The transformers shall be identified in accordance with MIL-STD-130, except that the serial number shall not be included. The markings shall be molded on the surface of the transformer. The following marking shall be included:

Manufacturer's Part No. _____ Secondary Amperes _____

Rating: Watt _____ Federal Stock No. _____

Primary Amperes _____

3.8 Workmanship. Each transformer shall be designed, constructed and finished in a manner that shall allow each transformer to meet the requirement specified herein and accomplish the purpose for which it is intended. The transformers, including all parts, components and assemblies, shall be clean of harmful extraneous materials such as loose, spattered, or excess solder, dirt, pits, or any other foreign material that might detract from the intended operation, function, or appearance of the transformer. This would include any particles that could loosen or become dislodged during the normal expected life of the transformer. There shall be no evidence of burns, abrading, or pinch marks in the insulation that could cause short circuits or leakage. Mating surfaces of each connector and cap shall be free from blemishes, holes, burrs, cracks, and rough spots. Particular attention shall be given to soldering, wiring, impregnation of coils, marking of parts and assemblies, and freedom of exposed parts from burrs and any other imperfection that might affect the transformers serviceability or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other acilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:08Z
Check the source to verify that this is the currentversion before use.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction transformers shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The preproduction transformers shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for tests shall be in accordance with MIL-STD-105, inspection level I.

4.4.2 Examination. Each transformer shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Tests.

4.4.3.1 Individual. Each transformer that has successfully completed the examination specified in 4.4.2 shall be tested as specified in 4.5.2.2.1. Failure of the test shall be cause for rejection.

4.4.3.2 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.2.1 through 4.5.2.4.8, except that only two transformers from a sample lot shall be subjected to the impact and vibration tests specified in 4.5.2.3.1 and 4.5.2.3.2, respectively. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The transformers shall be examined as specified herein for the following defects:

101. Components missing or not as specified.
102. Materials not as specified.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:08Z
Check the source to verify that this is the currentversion before use.

103. Weight in excess of that specified.
104. Identification marking missing, illegible or not as specified.
105. Workmanship not as specified.

4.5.2 Test.

4.5.2.1 Test conditions. Unless otherwise specified herein, tests shall be conducted at room temperature (approximately 25° C.). The secondary current electrical tests shall be conducted with a 60-cycle sine wave power source with primary current maintained at the values specified in table I.

4.5.2.2 Electrical characteristics.

4.5.2.2.1 Secondary current characteristics. The transformer shall be energized and the electrical characteristics determined after the transformer windings have stabilized. At each specified primary current and secondary load, the secondary current shall be within the limits specified in table I. Nonconformance to the electrical characteristics specified in table I shall constitute failure of this test.

Table I. Electrical Characteristics

Transformer rating (watts)	Primary amperes	Primary Power factor (minimum) percent	Efficiency (minimum) percent	Maximum Secondary volts (rms)	Secondary amperes	Load
1	2	3	4	5	6	7
30 to 45	6.6	95	80	-	6.53 to 6.67	1.15 ohms (45w lamp plus losses)
30 to 45	6.6	95	78	-	6.53 to 6.73	0.80 ohm (302w lamp plus losses)
30 to 45	6.6	-	-	-	6.6 to 7.1	Short circuited
30 to 45	6.6	-	-	25	0	Open circuited
30 to 45	5.3	-	-	-	5.2 to 5.4	0.96 ohm (45w lamp plus losses
30 to 45	.4	-	-	-	4.3 to 4.5	0.81 ohm (45w lamp plus losses

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:08Z
Check the source to verify that this is the current version before use.

Table I. Electrical Characteristics

1	2	3	4	5	6	7
30 to 45	3.8	-	-	-	3.7 to 3.9	0.72 ohm (45w lamp plus losses)
30 to 45	3.4	-	-	-	3.3 to 3.5	0.65 ohm (45w lamp plus losses)
200	6.6	95	90	-	6.53 to 6.67	4.82 ohms (200w lamp plus losses)
200	6.6	-	-	-	6.6 to 7.1	Short circuited
200	6.6	-	-	100	0	Open circuited
200	5.2	-	-	-	5.1 to 5.3	3.92 ohms(200w lamp plus losses)
200	4.1	-	-	-	4.0 to 4.2	3.22 ohms (200w lamp plus losses)
200	3.4	-	-	-	3.3 to 3.5	2.75 ohms (200 lamp plus losse
200	2.8	-	-	-	2.7 to 2.9	2.33 ohms (200w lamp plus losses)

4.5.2.2.2 Insulation resistance. The transformers that have passed the impact and vibration tests specified in 4.5.2.3.1 and 4.5.2.3.2, respectively, shall be subjected to a continuous 20-cycle test, in which one cycle shall consist of the sequence of operations specified in 4.5.2.2.2.1, 4.5.2.2.2.2, and 4.5.2.2.2.3. Mating connectors shall be installed in the three connectors of each transformer. The mating connectors shall not be removed before completion of the 20 cycles. If the mating connectors are removed, the tests shall be repeated so that each transformer and its connectors pass 20 continuous cycles. Measurements of insulation resistance shall be made with direct current. The insulation resistance shall be measured while the transformer and its connectors are immersed in a grounded tapwater bath. The voltage specified in table II shall be applied for 1 minute between each coil and ground with the other coil grounded. Zero and maximum readings of the test instrument shall be daily checked by immersing the high-voltage lead in water alongside a transformer and by suspending it in air. An insulation resistance less than the values specified in table II or any evidence of damage shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:08Z
Check the source to verify that this is the currentversion before use.

Table II. Insulation Resistance Values

Coil	DC test voltage	Minimum insulation resistance in megohms	
		Cold	Hot
Secondary	5,000	1,000	300
Primary	15,000	2,000	750

4.5.2.2.2.1 The transformers shall be operated for a minimum of 5 hours in air at room temperature with 6.6 amperes flowing in the primary and with the secondary open circuited.

4.5.2.2.2.2 At the end of the 5 hour period of 4.5.2.2.2.1, the transformers and connectors shall be immediately immersed in a grounded tapwater bath at room temperature to a depth of approximately 1 foot, and the insulation of each coil shall be measured within 3 minutes. The measurements shall be made with the connectors submerged in water with the bare ends of the mating connector leads out of the water.

4.5.2.2.2.3 The transformer and its connectors shall be soaked in water at room temperature for not less than 12 hours and the insulation resistance measurements shall be repeated.

4.5.2.2.3 Insulation resistance (alternate). Subject to the approval of the contracting officer, the following modifications may be made in the test specified in 4.5.2.2.2.

4.5.2.2.3.1 The test specified in 4.5.2.2.2.1 may be conducted by heating the transformers in an oven for not less than 2 hours, or by short circuiting either the primary or secondary coil and circulating slightly more than rated current through the other coil. The heating shall produce equivalent or greater temperatures in the coils, case, and leads of a transformer than the 5-hour open-circuit conditions.

4.5.2.2.3.2 Primary connectors of several transformers may be plugged directly together and the transformers heated, soaked, and tested as one transformer. The minimum acceptance insulation resistance for the combined primary coils of the group shall be not less than that specified for a single transformer. No connector in the group shall be unplugged until after the tests are completed.

4.5.2.3 Physical characteristics. Transformers which have passed the tests specified in 4.5.2.2.1 shall be used for the following impact and vibration tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:08Z
Check the source to verify that this is the current version before use.

4.5.2.3.1 Impact. The transformers shall be dropped twice from a height of 4 feet onto a hard, rigid surface, once in such a manner that they hit on a bottom corner or location where the most damage is likely to occur because of the core cutting into the case, and once in such a manner that they hit upon a side or location where the most damage is likely to occur to the windings. After being dropped, the transformers shall be subjected to the test specified in 4.5.2.2.1. Any evidence of damage to the case or a change of more than 1 percent in the results obtained in the test specified in 4.5.2.2.1 shall constitute failure of this test.

4.5.2.3.2 Vibration. With the transformer fastened to a vibration table to prevent slippage, test in accordance with MIL-STD-202, Method 201. Examine the transformer for damage, and test in accordance with 4.5.2.2.1. Any damage to the transformer or a change of more than 1 percent in the results obtained in the test specified in 4.5.2.2.1 shall constitute failure of this test.

4.5.2.4 Environmental characteristics. Unless otherwise specified herein, the transformer shall be subjected to the following tests conducted in accordance with MIL-STD-810.

4.5.2.4.1 Temperature rise. The temperature rise of the transformer windings not subjected to the impact test specified in 4.5.2.3.1 shall be determined by the resistance method. Nonconformance to 3.5.3.1 shall constitute failure of this test.

4.5.2.4.2 High temperature. The transformers shall be subjected to high temperature in accordance with Method 501, Procedure I. Any damage to the transformer or failure to operate as specified shall constitute failure of this test.

4.5.2.4.3 Low temperature. The transformers shall be subjected to low temperature in accordance with Method 502, Procedure I. Any damage to the transformer or failure to operate as specified shall constitute failure of this test.

4.5.2.4.4 Sunshine. The transformer shall be subjected to solar radiation in accordance with Method 505, Procedure I. Any damage to the transformer or failure to operate as specified shall constitute failure of this test.

4.5.2.4.5 Moisture. The transformer shall be subjected to high humidity and heat in accordance with MIL-STD-202, Method 106. After the high humidity test and the drying period specified in Method 106, the transformers shall be tested as specified in either 4.5.2.2.1 or 4.5.2.2.2. Any damage to the transformer, failure to operate as specified, or insulation resistance values less than those specified in table II shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:08Z
Check the source to verify that this is the currentversion before use.

4.5.2.4.6 Fungus. The transformers shall be subjected to fungi in accordance with Method 508. Any damage to the transformer or failure to operate as specified shall constitute failure of this test.

4.5.2.4.7 Oil. The transformer, including the mating connectors, shall be tested as specified in the Method 4221 of FED. STD. No. 228, except that a reflux condenser is not required and the container shall be a tub filled with oil. One or more transformers may be tested at the same time provided they do not touch each other or the surfaces of the container. After the transformer has been removed from the cooling liquid, it shall be tested as specified in 4.5.2.2.1. Failure to operate as specified in 4.5.2.2.1 or damage to the transformer shall constitute failure of this test.

4.5.2.4.8 Acid and alkali. Four transformers, including their mating connectors, shall be tested for resistance to the acid and alkali solutions specified in Method 7011 of FED. STD. No. 228. One transformer shall be immersed in each of the solutions for a period of 46 hours. At the end of the exposure period, each transformer shall be removed from the solution, rinsed for 15 seconds in running water, shaken to remove any loose water, and any adhering surface water removed by blotting lightly with a clean absorbent cloth. Each transformer shall then be tested for the electrical characteristics specified in 4.5.2.2.1 and examined for damage. Failure to operate as apecified in 4.5.2.2.1 or evidence of damage, including acid and alkali absorption, shall constitute failure of this test.

4.6 Inspection of preparation for delivery. The preservation, packaging and packing shall be examined and tested in accordance with MIL-E-17555. The marking shall be examined to determine if any marking is missing, illegible, incorrect, or incomplete in accordance with MIL-E-17555. Presence of one or more defects in the marking shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. The transformers shall be preserved, packaged, packed, and marked in accordance with MIL-E-1755 The level of preservation and packaging shall be level A or C and the level of packing shall be level A, B, or C as specified (see 6.2). For military activities, fiberboard containers are not acceptable for level A packing.

6. NOTES

6.1 Intended use. The transformers are intended for use in both permanent and advance airport lighting series-circuit installations.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of transformer required (see 1.2).

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:08Z
Check the source to verify that this is the currentversion before use.

(c) Time frame required for submission of preproduction model and number of models required (see 3.2).
(d) Levels of preservation, packaging and packing required (see 5.1).

6.3 Information figure. Figure 1 shows a type of transformer that has been found acceptable; however, the figure is included for illustration only and is not intended to preclude the furnishing of other transformers which conform to this specification.

6.4 Definitions. The following definitions shall apply throughout this specification.

6.4.1 Damage. Damage is:

(a) Deformation, loosening, breakage, deterioration or water leakage of the transformer, including any component or part.
(b) Reduction of insulation resistance below the values specified in table II.
(c) Corrosion of any hardware.
(d) Any condition resulting in malfunctioning of the transformer.

6.5 Preproduction model. Any changes or deviations of production transformers from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish transformers conforming to this specification.

Custodian:

Army - ME

Review activity:

Army - EL

Preparing activity:

Army - ME

Project No. 5950-A107

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:08Z
Check the source to verify that this is the currentversion before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-T-52497A(ME) - TRANSFORMERS, POWER, ISOLATION: SERIES-CIRCUIT, AIRPORT LIGHTING, 6.6/6.6 AMPERES

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 64 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:08Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
U. A. ARMY MOBILITY EQUIPMENT COMMAND
RESEARCH, DEVELOPMENT AND ENGINEERING DIRECTORATE
ATTN: AMSME-RZD
FORT BELVOIR, VIRGINIA 22060

OFFICIAL BUSINESS

POSTAGE AND FEES PAID

DEPARTMENT OF THE ARMY
U. S. ARMY MOBILITY EQUIPMENT COMMAND
RESEARCH. DEVELOPMENT AND ENGINEERING DIRECTORATE
ATTN: AMSME-RZD
FORT BELVOIR, VIRGINIA 22060

FOLD

D—36,485—Ft

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:08Z
Check the source to verify that this is the current version before use.